

Junwen Liao

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Research Profile

Ph.D. candidate in mathematics working at the intersection of AI for Science, computational structural biology, and inverse problems. I develop methods that couple frozen protein structure priors with limited cryo-EM observations through differentiable physical forward models and target-specific test-time adaptation. My current focus is experimentally guided protein diffusion models and scalable refinement of large biomolecular complexes, with particular attention to reproducibility, identifiability, and computational efficiency.

Education

2024–Present	Ph.D. Candidate in Mathematics Advisor: Prof. Chenglong Bao.	Qiuzhen College, Tsinghua University
2020–2024	B.S. in Statistics Advisor: Prof. Zhiguo Wang. GPA: 3.89/4.00.	School of Mathematics, Sichuan University

Selected Research Experience

2024.06–Present	Multimodal Integration of Cryo-EM and Protein Structure Priors Advisor: Prof. Chenglong Bao - Developed CoCoFold (<i>Communications Chemistry</i>, 2026; co-first author), integrating raw cryo-EM particles, poses, and CTF parameters with AlphaFold through differentiable projection and Fourier ring correlation supervision. - On MSP-1 with 1.1k particles, direct 2D supervision achieved a TM-score of 0.97 and RMSD of 1.82 Å, compared with 0.86 and 4.91 Å for 3D-map supervision; robustness was further evaluated across 10 datasets and 26 missing-view cases. - Extended this direction in CoCoFold2, a preprint on scalable latent refinement of frozen diffusion-based protein structure priors using limited-particle cryo-EM observations. - CoCoFold2 freezes a Protenix-v1/AF3-style diffusion prior, fixes the diffusion sampling state, and optimizes target-specific latent corrections against differentially rendered particle observations; the current implementation also supports component-parallel refinement for larger biomolecular systems. - Defined explicit applicability boundaries for particle-guided refinement: the framework relies on a sufficiently accurate initial prior, informative observations, and reliable upstream pose/CTF estimates. CoCoFold Paper / CoCoFold Code / CoCoFold2 Project Page / CoCoFold2 Code.
2023.08–2024.03	Deep Learning for Compressed-Sensing Computational Imaging Advisor: Prof. Chenglong Bao - Built a neural reconstruction framework for simulated integrated-circuit imaging by unrolling filtered back-projection and treating projection angles as learnable parameters. - Benchmarked the framework against traditional reconstruction algorithms and neural-network baselines, obtaining more accurate reconstruction with a smaller parameter count. - Awarded Outstanding Undergraduate Thesis, Sichuan University.

2022.07–2023.04

Chemical Identification from Terahertz Absorption Spectra

Advisors: Prof. Zhiguo Wang and Prof. Bo Wang

- Constructed a standardized terahertz absorption-spectrum database by modeling first- and second-order statistical moments under varying experimental conditions.
- Developed and benchmarked a maximum-a-posteriori quadratic classifier with shrinkage against peak-based, conventional machine-learning, and neural-network methods.
- Published in *Sensors and Actuators B: Chemical*, 2023. [Paper](#).

Preprints

1. **Junwen Liao**, Mingxu Hu, and Chenglong Bao. “CoCoFold2: scalable latent refinement of diffusion-based protein structure predictions from limited-particle cryo-EM data.” *LangTaoSha Preprint Server*, 2026. [Project Page](#) / [Code](#).

Selected Publications

1. **Junwen Liao***, Dihan Zheng*, Hui Zhang*, Linfeng Zhang, Mingxu Hu, and Chenglong Bao. “Fine-tuning AlphaFold with limited cryo-EM observations.” *Communications Chemistry*, 2026. [Code](#).
2. **Junwen Liao**, Bo Wang, Zhiguo Wang, and Liguu Zhu. “Amino-acid classification based on terahertz absorption spectroscopy with Gaussian process and maximum likelihood.” *Sensors and Actuators B: Chemical*, 2023.

* Equal contribution.

AI and Research Computing

Scientific AI	AlphaFold-Multimer, Protenix-v1/AF3-style diffusion models, differentiable Gaussian rendering, and attention-based models.
Research engineering	Python, PyTorch, NumPy/SciPy, distributed computing, Slurm, Git, LaTeX, and reproducible experiment pipelines.
AI-assisted workflow	LLMs and coding agents for literature triage, codebase navigation, prototyping, unit testing, debugging, and documentation; outputs are verified through controlled experiments, code review, and reproducible logs.

Awards and Honors

- Outstanding Graduate of Sichuan Province.
- Outstanding Student, Sichuan University.
- First-Class Comprehensive Scholarship, Sichuan University.
- Meritorious Winner, Mathematical Contest in Modeling (MCM/ICM).
- First Prize, Provincial Level, Chinese Mathematics Competitions.

Teaching Experience

Fall 2026	Teaching Assistant, Linear Algebra Assisted with course instruction and student support.	Tsinghua University
Fall 2024	Teaching Assistant, Linear Algebra Assisted with course instruction and student support.	Tsinghua University